Concept Review

Rotary Encoders

Why Use Rotary Encoders?

Rotary encoders measure rotational motion. They are used to monitor position and speed in applications such as industrial machinery, automotive vehicles, and robotics. Their ability to precisely track position allows for fine movement control in surgical robots, aerospace applications, and CNC machines. Compared to rotary potentiometers, encoders offer higher precision, long-term reliability, and unlimited movement range, but require more complex implementation in software to track and process their inputs.

Rotary Encoders

Rotary encoders measure rotational position, speed, and direction. There are many types of encoders but one of the most common is the rotary incremental optical encoder, shown in Figure 1. Unlike rotary potentiometers, which measure absolute position, incremental encoders measure relative position.

Figure 1: US Digital incremental rotary optical shaft encoder

Operating Principle

The encoder consists of a code disc marked with a radial pattern of evenly spaced slots that rotates with the encoder shaft, as well as a light source and photosensors, which are fixed to the assembly. When the encoder shaft and code discs are rotated, the photo sensors pick up the pattern of pulses generated by light shining through the slots. Two photosensors generate the signals for the A and B channels, as shown in Figure 2.

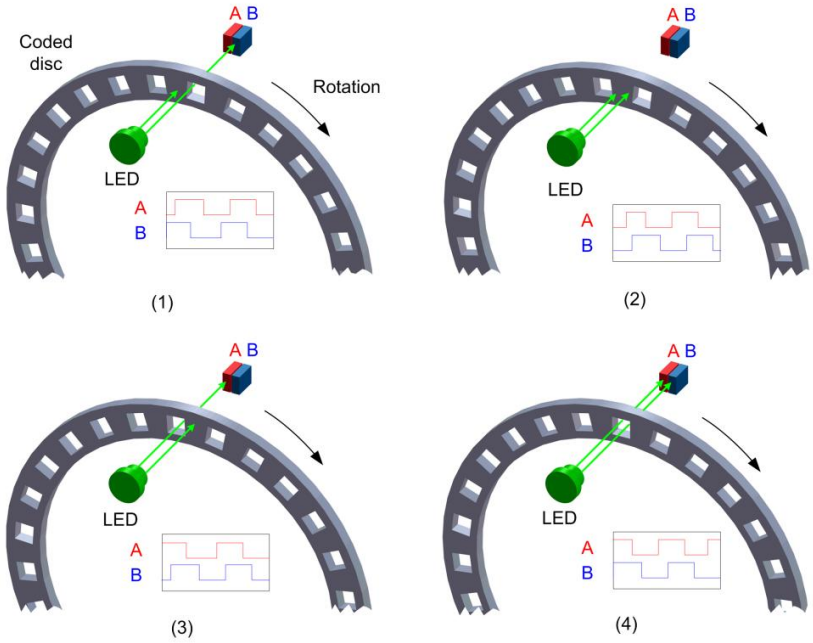

Figure 2: Optical incremental encoder signals

Some encoders have an additional channel where an index pulse is triggered once for every full rotation of the disc, which can be used for calibrating or homing a system. The A and B signals that are generated as the shaft rotates are used in a decoder algorithm to generate a count. The resolution of the encoder depends on the coding of the disc and the decoder. For example, a single encoder with 512 lines on the disc can generate a total of 512 counts for every rotation of the encoder shaft. However, in quadrature decoder as depicted in Figure 2, the number of counts (and thus its resolution) quadruples for the same line patterns and generates 2048 counts per revolution. This can be explained by the offset between the A and B patterns: Instead of a single strip being either on or off, now there is two strips that can go through a variety of on/off states before the cycle repeats. This offset also allows the encoder to detect the directionality of the rotation, Figure 3 demonstrates X1 decoding (only using the rising edges of the A signal), X2 decoding (using the rising and falling edges of the A signal) as well as X4 decoding (using the rising and falling edges of both A and B signals, otherwise known as quadrature decoding).

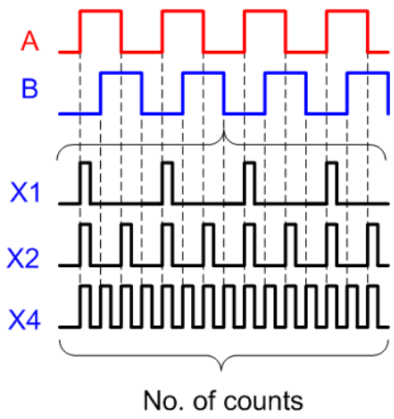

Figure 3: X1, X2 and X4 decoding with optical encoders

Absolute Encoders

Absolute encoders function slightly differently than incremental encoders to measure absolute angular position instead of the relative position. Absolute encoders provide a unique code for each shaft position on the encoder disc. The resolution is determined by its number of code rings, which corresponds to the number of bits required to represent the on (1) and off (0) pattern in each segment. An n-bit absolute encoder has n code rings, which can encode 2^n segments, shown in Figure 4 below.

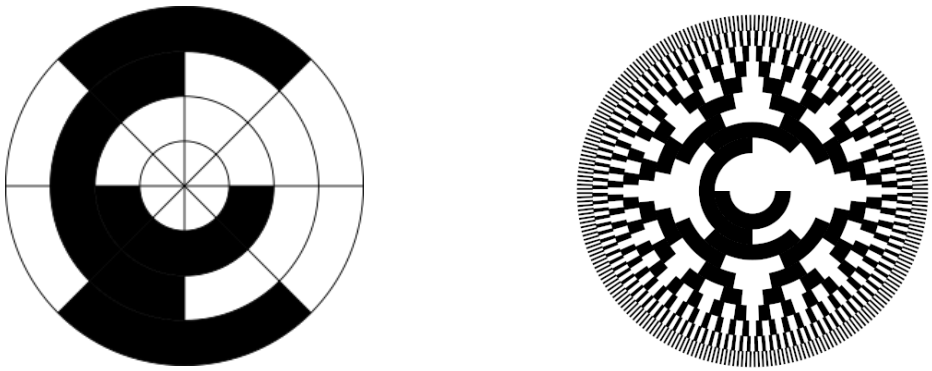

Figure 4: Code discs for a 3-bit (left) and 10-bit (right) absolute encoder

The benefit of using an absolute encoder is that the position measurement does not require the initial position of the shaft to be known. As such, absolute encoders are well-suited for applications requiring precise position sensing immediately when it is powered on, such as the measurement of joint positions in a robotic arm. Incremental encoders are used in applications where the initial position is fixed or it is more important to know relative position, such as conveyor belts or motor speed control.

Datasheet and Considerations

Consult the device datasheet and consider the needs of the specific application when selecting an encoder. An important specification to consider is the resolution, which determines the minimum movement angle that can be detected by the encoder. Encoders with higher resolutions are able to detect smaller changes, which may be useful for systems that depend on measurements of very small movements, or where movement occurs at slower speeds.

This document focused on optical rotary incremental encoders, but it is important to consider that other types of encoders may be more appropriate for certain position-sensing applications. Linear encoders measure position along a single axis, and come in both incremental and absolute variations. Optical encoders, as described in this document, work by emitting and detecting light, but other types of encoders, such as magnetic encoders, work by measuring other types of signals.

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